

Intro2SE – Design – GroupID

Project Name

*The student team is required to complete the **Design Document** for the assigned course project, following the attached template.*



Software Engineering Department
Faculty of Information and Technology
VNU-HCMUS

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Software Design

Objectives

This document focus on the following topics:

- ✓ Completing the Software Design Document with the following contents:
 - Conceptual Model
 - Architectural Design
 - Data Design
 - User Interface Design
- ✓ Understanding the Software Design Document.
- ✓ This document will be used as input for AI tools to verify the quality of subsequent project artifacts.

All project artifacts must remain consistent and synchronized.

For example, if the project proposal is modified during this phase, a new version of the proposal must be documented.

By the conclusion of the project, all versions of every artifact must be submitted to demonstrate the evolution of your work.

1 Member Contribution Assessment

StudentID 1 – Fullname 1 – Contribution (%)

- Task Evidence: Provide a comprehensive list of tasks managed via Jira (or a similar project management tool). For each entry, include: Task Details: Task name and a descriptive summary of the work performed. Visual Evidence: Clear screenshots that explicitly show the Task Name, Assignee, Status, Assigned Date, and Completion Date.

StudentID 2 – Fullname 2 – Contribution (%)

- Task Evidence

Each task must be assigned to and completed by only one student.

The sum of all student contributions is less than or equal to 100%.

2

Conceptual Model

[Present a diagram illustrating the semantic entities within the software. This section can be created using an Extended Entity-Relationship (EER) model generated by **MySQL Workbench** or **PowerDesigner**]

3

Architectural Design

1.1 Architecture Diagram

[Present a **system decomposition tree diagram**, indicating the components of the system]

[Present an overall system architecture diagram, illustrating the relationships between the **MAIN components** identified in the system decomposition tree]

[If there are any special aspects of the architecture, such as the application of Design Patterns, the use of Client-Server/Tier/MVC architecture, support for a plug-in mechanism, etc., these should be clearly presented to demonstrate this]

1.2 Class Diagram

[The class diagram should clearly show:

- The names of the object classes.
- The relationships between the object classes (relationship type, cardinality).

It is not strictly necessary to list the attributes and methods of each object class in the class diagram. However, this information should be included if the diagram is not too large]

1.3 Class Specifications

[Students should select and present the specifications of a few (9-10) of the most important classes.]

1.3.1 Class C1

[Which class does this class inherit from (if any)]

[Describe the list of attributes (which attributes are inherited – if any, the attribute's visibility: **public**, **protected**, or **private**, etc.)]

Seq	Property	Modifier	Constraint	Description

[List of the main methods/operations]

Seq	Operation	Modifier	Constraint	Description

4

Data Design

1.4 Data Diagram

[Draw the data diagram of the system, identifying the data components that need to be stored and illustrating the relationships between them]

1.5 Data Specification

[If using a Database Management System (DBMS), describe the data tables, the information for each data column including attribute name, data type and value constraints, key constraints..., and explain the attribute]

[If using XML/JSON or a self-defined structured file to store data, the file structure, attribute information, data type, and value constraints need to be specifically described. An example of the content for the information storage file should be included]

5

User Interface and User Experience Design

1.6 Screen Diagram

[Draw a screen diagram illustrating the relationships and transitions between the screens]

[List the screens]

Seq	Screen	Description

1.7 Screen Specifications

[Students should select and present the specifications of a few (5-6) of the most important screens. For the remaining screens, only the user interface design needs to be drawn]

1.7.1 Screen "A"

[Clearly describe the presentation format and the handling of each event within the screen. Include a design image of each screen]

1.7.2 Screen "B"

[Clearly describe the presentation format and the handling of each event within the screen. Include a design image of each screen]

6

AI Usage Declaration

Based on the provided AI Usage Guideline document, teams must submit their AI usage declaration here. Students are encouraged to leverage AI tools creatively and effectively in this project.

7 Presentation

Teams are required to record presentation videos of their work using the provided template.

Every team member must participate in the presentation, specifically covering the sections their contributed to the project.

Videos must not exceed 30 minutes.

Please upload your videos to Youtube (set to either Unlisted or Public).

Provide the Youtube links the the field below.